



BLOC 3 – Machine Learning

North Face project

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Optimiser l'offre produit e-commerce grâce au Machine Learning non supervisé

Structurer l'offre produit

Regrouper les articles similaires via analyse de texte (descriptions produits)
Identifier des familles homogènes pour simplifier la navigation et le merchandising

Améliorer l'expérience utilisateur

Proposer des recommandations plus pertinentes entre produits similaires
Faciliter les parcours d'achat grâce à une segmentation plus intuitive

Appuyer les décisions marketing et commerciales

Mieux comprendre la diversité de l'offre
Identifier les segments sous-exploités ou à fort potentiel

Explorer des méthodes avancées de Machine Learning

Utiliser le clustering (DBSCAN) pour regrouper les produits sans a priori
Appliquer la réduction de dimension (LSA) pour extraire les thèmes dominants
Expérimenter des techniques de topic modeling pour enrichir la compréhension produit



Données Produits e-commerce (NLP)

- Base de données très simple : 500 produits textile, chacun identifié par un ID unique
- Une seule colonne exploitable : description, combinant nom du produit et description marketing

```
1 id,description
2 1,"Active classic boxers - There's a reason why our boxers are a cult favorite - they keep their cool, especially in sticky situations. The quick-drying, lightweight u
3 2,"Active sport boxer briefs - Skinning up Glory requires enough movement without your boxers deciding to poach their own route. The form-fitting Active Sport Boxer Br
4 3,"Active sport briefs - These superbreathable no-fly briefs are the minimalist's choice for high-octane endeavors. Made from a blend of fast-wicking, quick-drying 93%
5 4,"Alpine guide pants - Skin in, climb ice, switch to rock, traverse a knife-edge ridge and boogie back down - these durable, weather-resistant and breathable soft-she
6 5,"Alpine wind jkt - On high ridges, steep ice and anything alpine, this jacket serves as a true ""best of all worlds"" staple. It excels as a stand-alone shell for bl
7 6,"Ascensionist jkt - Our most technical soft shell for full-on mountain pursuits strikes the alpinist's balance between protection and minimalism. The dense 2-way-str
8 7,"Atom - A multitasker's cloud nine, the Atom plays the part of courier bag, daypack and carry-on. Its teardrop shape provides the support of a backpack by positioning
9 8,"Print banded betina btm - Our fullest coverage bottoms, the Betina fits highest across the hips with a slightly scooped, lined front. Made from a blend of 82% nylon
10 9,"Baby micro d-luxe cardigan - Micro D-Luxe is a heavenly soft fabric with down-to-earth applications. This cardigan is made from a quick-drying, durable 4.6-oz 100%
11 10,"Baby sun bucket hat - This hat goes on when the sun rises above the horizon, and stays on when raindrops start falling. Its made from an ultra-durable 4-ply, 4.2-o
```



Men's Antora Jacket

Waterproof, windproof jacket for outdoor adventure ⓘ

\$110.00

96% Customer Recommended

★★★★★ [1131 Reviews](#)

Description

Style NF0A7QEY4H0

When the weather turns, we turn to the Antora. With practical design elements inspired by our iconic Mountain Jacket, this highly versatile rain shell sets the new standard for everyday reliability. Engineered for an easy transition from rainy sidewalks to on-trail downpours, it's built using our fully recycled waterproof, windproof DRYVENT™ fabric. This is town-to-trail coverage at its best, focused on the details that matter most when the clouds roll in. Wear it all day long or pack it just in case—the Antora is always ready.





Techniques de nettoyage et préparation

- Mise en minuscule
- Exclusion des caractères spéciaux
- Tokenisation / Lemmatization
- Exclusion des tokens nombres
- Exclusion des tokens de 2 lettres
- Exclusion des tokens Stop_Words

```
0 [active, classic, boxer, reason, boxer, cult, ...  
1 [active, sport, boxer, brief, skin, glory, req...  
2 [active, sport, brief, superbreathable, fly, b...  
3 [alpine, guide, pant, skin, climb, ice, switch...  
4 [alpine, wind, jkt, high, ridge, steep, ice, a...  
Name: complete, dtype: object
```



Regroupement de produits similaires

- Objectif : 10 à 20 groupes d'au moins 6 produits en minimisant les outliers
- Méthode de clusterisation DBSCAN en optimisant les paramètres
 - > 12 clusters réels
 - > Un 13^e cluster regroupant les outliers (91 références)
 - > Taille médiane des clusters réels : 12 références

	id	title	description	clean_token	best_dbscan_cluster
0	1	Active classic boxers	There's a reason why our boxers are a cult fav...	active classic boxer reason boxer cult favorit...	3
1	2	Active sport boxer briefs	Skinning up Glory requires enough movement wit...	active sport boxer brief skin glory require mo...	-1
2	3	Active sport briefs	These superbreathable no-fly briefs are the mi...	active sport brief superbreathable fly brief m...	3
3	4	Alpine guide pants	Skin in, climb ice, switch to rock, traverse a...	alpine guide pant skin climb ice switch rock t...	0
4	5	Alpine wind jkt	On high ridges, steep ice and anything alpine,...	alpine wind jkt high ridge steep ice alpine ja...	0



Système de recommandation produit

- Exemple d'utilisation pratique des clusters
- Entrer un **ID référence**, et trouver 5 autres produits faisant partie du même cluster via une simple boucle Python -> système de recommandation

```
You selected product ID #47
Title: Girl's cotton tank dress
Description: This soft cotton dress feels breezy and cool, just like a day at the beach. Made of a lightweight 5-oz 100% organic cotton jersey-knit fabric that breathes naturally

This product is part of cluster #10.
Here are 5 similar items:

+ ID #250 - Simply organic tee
Description: The Simply Organic Tee survives tough scrapes year after year, making it a perennial favorite. Made from soft organic cotton, the shirt layers beautifully and has form

+ ID #346 - Versatiliti tee
Description: A free-moving, breathable and famously resilient tee-shirt, the Versatiliti is designed for the perfect blend of town and country. Its soft 5.5-oz 62% organic cotton i

+ ID #311 - Astrid wrap
Description: A casual, elegant wrap to layer over your favorite dress - or to throw on over a tee and jeans. The short-sleeved Astrid Wrap is made of our lightest weight organic co

+ ID #162 - Astrid tank
Description: Conjure up Cardiff's shouldery waves with the Astrid Tank. The unbelievably soft, strappy tank has a flattering scoop neck and is made from our lightest weight organic

+ ID #298 - Versatiliti tank
Description: Vivid colors and delicate details enliven this season's crop of Versatiliti Tanks. Made of a lightweight blend of 62% organic cotton/38% Tencel that line-dries over ni
```



Analyse sémantique – Identification des thèmes

Méthodologie utilisée :

- ▶ TF-IDF : conversion des textes en vecteurs pondérés
- ▶ Truncated SVD (LSA) : réduction de dimension et extraction des composantes latentes
- ▶ Chaque composante $\approx$ un thème dominant dans les descriptions

Thèmes identifiés

● Topic 0: pocket | organic | cotton | recyclable | polyester | common | program | thread | recycling | fabric
● Topic 1: organic | cotton | shirt | print | ring spun | phthalate | pvc | ink | tape | screen
● Topic 2: merino | odor | wool | control | gladiodor | natural | capilene | skin | wash | knit
● Topic 3: inseam | organic | cotton | button | short | canvas | jean | fly | welt | size
● Topic 4: merino | wool | wash | machine | slow | lie | naturally | resist | cold | core
● Topic 5: nylon | sun | spandex | upf | coverage | protection | rise | print | blend | usa
● Topic 6: spandex | coverage | tencel | blend | line | cotton | organic | hip | bra | jersey
● Topic 7: strap | mesh | polyurethane | denier | pocket | compartment | coating | shoulder | hold | access
● Topic 8: sun | upf | protection | collar | recycled | polyester | reflective | rashguard | runshade | sleeve
● Topic 9: fleece | spandex | wind | stretch | brush | warmth | handwarmer | interior | trim | waistband
● Topic 10: photo | poster | outside | brighten | catalog | clothier | dimensions | excited | exclusively | hope
● Topic 11: fleece | microfiber | polyester | wader | faced | trim | hydrostorm | proprietary | zip | collar
● Topic 12: microfiber | stretch | polyester | hydrostorm | proprietary | weave | gravel | wader | heat | reflective
● Topic 13: construction | canvas | heat | toe | sock | foot | logo | underarm | reflective | denim
● Topic 14: jean | coverage | denim | wash | fleece | styling | line | microfiber | btm | improve
● Topic 15: denim | jean | fly | zip | styling | knit | right | loop | fleece | knee
● Topic 16: neck | rashguard | rib | chafe | label | tender | flat | spandex | screen | size
● Topic 17: reflective | logo | transfer | heat | runshade | fleece | moisture | wicke | knit | glare
● Topic 18: support | neck | insole | supplex | fleece | bra | water | baggie | elasticize | durable
● Topic 19: barely | bra | support | moisture | recycle | seamless | waistband | spandex | shoulder | removable



Bilan du projet – Analyse Produit e-commerce

Résultats obtenus

Identification de groupes de produits similaires via analyse textuelle
Extraction de thématiques clés à partir des description marketing
Méthodologie scalable, automatisable, et sans supervision humaine
Piste concrète pour structurer l'offre et guider les recommandations produit

Limites actuelles

Impact opérationnel limité en l'état
▶ Résultats difficilement exploitables sans croisement avec d'autres sources
Volume de données restreint (500 produits)
▶ Limite la robustesse des regroupements



Recommandations ciblées pour progresser

- **Coupler avec des données comportementales** (navigation, clic, ventes, taux de conversion...) pour affiner la pertinence business
- **Etendre le périmètre** à un catalogue produit plus large pour renforcer l'analyse
- **Elargir l'analyse** avec des données descriptives (prix, collection...) et visuelles (images, couleurs, styles...)
- **Enrichir avec une couche de validation métier** (experts, merchandising...) pour lier au commerce physique





Merci !

